

Amirul Hakim

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EDUCATION

Bachelor of Engineering in Mechanical Engineering | Universitas Pancasila

September 2022 – July 2026

Cumulative GPA: 3.72/4.00

Undergraduate Thesis: Effect of Feed Rate in Micro-Milling on Burr Height and Dimensional Accuracy of Titanium I-Plate Implants; experimental machining of Commercially Pure Titanium with micro-milling method and analyzed optical dimensional and burr-height measurement using digital microscopy and image processing software ImageJ. The research showed that

EXPERIENCE

Manufacturing Engineering Intern | PT Defa Angkasa Utama

August 2025 – October 2025

- Analyzed end-to-end sheet-metal HVAC louver production and mapped 31 operation/inspection stages across an approximately 300-minute production cycle using an Operation Process Chart (OPC) and helped with standardizing process with Standard Operating Procedure (SOP), increasing work efficiency by 22%.
- Identified manual welding setup and component positioning as repeatability constraints and proposed a modular welding-jig concept for configurable louver sizes potentially saving welding process time up to 60%.

Maintenance and Instrumentation Intern | PT Timah Industri

September 2024 – July 2025

- Developed an Arduino/ESP32-based monitoring system for chiller system, acquiring field data through RS-485/Modbus RTU and publishing telemetry via MQTT into MySQL-backed dashboard for real-time and historical monitoring of flow, temperature, and related operating data.
- Assisted with design and implementation of a monitoring/control pH and Total Dissolved Solids (TDS) system based on Programmable Logic Controller for a Water Treatment Plant (WTP), integrated into centralized monitoring system, reducing potential total down time by 40%.
- Contributed to MEP planning and design with Autodesk AutoCAD and Revit for a production control-room expansion.

Mechanical Drafter | PT Jordyla Mitra Sejati

July 2023 – May 2024

- Produced and revised 2D manufacturing drawings and 3D CAD models for machined and fabricated components used in heavy-machinery and automotive applications.
- Prepared cutting profiles, part layouts, BOMs, drawing revisions, geometric dimensioning and tolerancing (GD&T), CNC programs, and other production documentation where applicable.
- Performed reverse engineering of undocumented components and coordinated drawing revisions with production personnel across machining, laser cutting, bending, welding, and assembly processes.
- Maintained drawing files, revisions, part numbers, and other technical documentation.

PROJECTS

Automated Industrial System for Water Treatment Plant

- Developed a PLC-based multi-pump control system with automatic duty/standby sequencing, tank-level control, pH level control, motor permissives and interlocks, fault handling, alarm management, and manual/automatic operating modes.
- Designed the HMI monitoring interface for pump status, level indication, pH level, alarms, operating commands, and equipment conditions, and simulated normal and abnormal operating scenarios to verify control sequences and equipment-protection logic.

Modular Roller Conveyor Mechanical Design

- Designed a modular non-powered roller conveyor for manual material handling, incorporating straight and curved conveyor sections, structural frames, adjustable support legs, connectors, end stoppers, and bearing-supported roller assemblies.
- Developed the complete 3D CAD assembly, individual component models, BOM, subassemblies, and five-sheet production drawing package, including sectional/detail views and manufacturing dimensions for fabrication and assembly.

Industrial Control Room Expansion – Mechanical & Facility Design

- Developed AutoCAD drawings for the expansion of an industrial production control room at PT Timah Industri, including floor layout, elevations, sections, structural framing, and integration with the existing facility.
- Specified and documented structural and architectural components including H-beam/UNP framing, EPS sandwich panels, aluminium window framing, glazing, flooring, dimensions, and installation geometry for fabrication/construction coordination.

CERTIFICATIONS

Certified SOLIDWORKS Professional (CSWP) | Dassault Systèmes

September 2024

Certificate of Competency in 3D CAD Drafting | BNSP

July 2024 – July 2027

Siemens PLC & HMI Programming Specialization | Packt, Coursera

March 2026

SKILLS

CAD & Mechanical Design: SolidWorks, AutoCAD, Autodesk Inventor, 3D modeling, assembly design, manufacturing drawings, detail drawings, BOM, GD&T, tolerance specification, reverse engineering.

Fabrication & Manufacturing: sheet-metal fabrication, laser cutting, press-brake bending, machining, MIG welding, jig & fixture design, CNC manufacturing, production documentation.

Additional Engineering: Revit, ANSYS/FEA/CFD, PLC/HMI, instrumentation, Modbus, MQTT..

Language: Indonesian (Native), English (Professional Working Proficiency), German (Intermediate Level)